Explanatory Data Analysis

Environmental Drivers of Purple Urchin Gonad Index in Central California Kelp Forests

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1. Introduction and Research Question

Kelp forests are among the most productive and biodiverse marine ecosystems on the Pacific coast, providing critical habitat structure and trophic support for a wide range of species. Within these ecosystems, purple sea urchins (*Strongylocentrotus purpuratus*) have a critical ecological role as kelp grazers. Their grazing behavior can either maintain healthy kelp forests or, under certain conditions, contribute to the formation of urchin barrens. These barren states are characterized by high urchin densities, usually dominated by crustose-coralline algae, and little to no macroalgae cover, representing a fundamentally different and less productive ecosystem state. Gonadosomatic index (GI), the ratio of gonad mass to total body mass, serves as an indicator of urchin condition and reproductive potential. As kelp forest restoration efforts expand along the California coast in response to widespread kelp decline, identifying which habitat characteristics predict urchin gonad condition is essential for prioritizing restoration sites and evaluating intervention success. This project applies a Bayesian principal component regression approach to identify key habitat predictors of purple urchin gonad index across survey sites in central California, with the aim of informing evidence-based kelp forest restoration.

Research Question

To what extent do biological and environmental variables predict **sea urchin gonad index** across kelp forests and urchin barrens, and which variables are the strongest predictors of gonad index?

2. Load Relevant Packages

```
library(tidyverse) #data wrangling (includes other packages)
library(dplyr) #data wrangling
library(ggplot2) #plotting
library(brms) #Bayesian statistics
library(bayesplot) #posterior probability distributions
library(factoextra) #variance plot (scree plot)
library(tidybayes) #helps convert Bayesian modeling for ggplot2 and tidyverse
library(patchwork) #put plots together
```

3. Data Wrangling

The data for this project came from a Monterey Bay Aquarium kelp forest recovery research program. From 2024-2025, data was collected during the summer field seasons. Data collected included SCUBA monitoring surveys (recording species density, kelp cover and topography) and purple sea urchin collection from 12 sites (with multiple zones within sites) around Carmel Bay and the Monterey Bay peninsula. Sea urchins were dissected in the lab to calculate mean gonadosomatic index for each zone within sites.

The raw dataset contains ~80 variables with 98 observations. To simplify the dataset for more accurate analysis, we selected 20 variables that we deemed as most relevant to our research question.

```
#raw data
predictors_raw <- readRDS("quad_build3.rds")

#select relevant columns and delete rows with NA values
predictors <- predictors_raw %>%
   mutate(juveniles = macj + nerj + ptej + lsetj + eisj,
         n_macro_plants_m2 = n_macro_plants_20m2/20,
         macro_stipe_density_m2 = macro_stipe_density_20m2/20,
         adults = (density20m2_ptecal + density20m2_eisarb + density20m2_nerlue + density20m2
         marine_snails = tegula_densitym2 + pomaulax_densitym2,
         density_cancer = density20m2_cancer_spp/20,
         density_lamstump = density20m2_lamstump/20,
         density_macstump = density20m2_macstump/20) %>%
  dplyr::select(mean_gi, n_macro_plants_m2, macro_stipe_density_m2, adults,
juveniles, density_cancer, density_lamstump, density_macstump, relief_cm, risk_index, purple_
  na.omit()
```

```
#final dataset for project
predictors2 <- predictors %>%
  dplyr::select(mean_gi, cov_mac_holdfast_live, macro_stipe_density_m2, n_macro_plants_m2, co
```

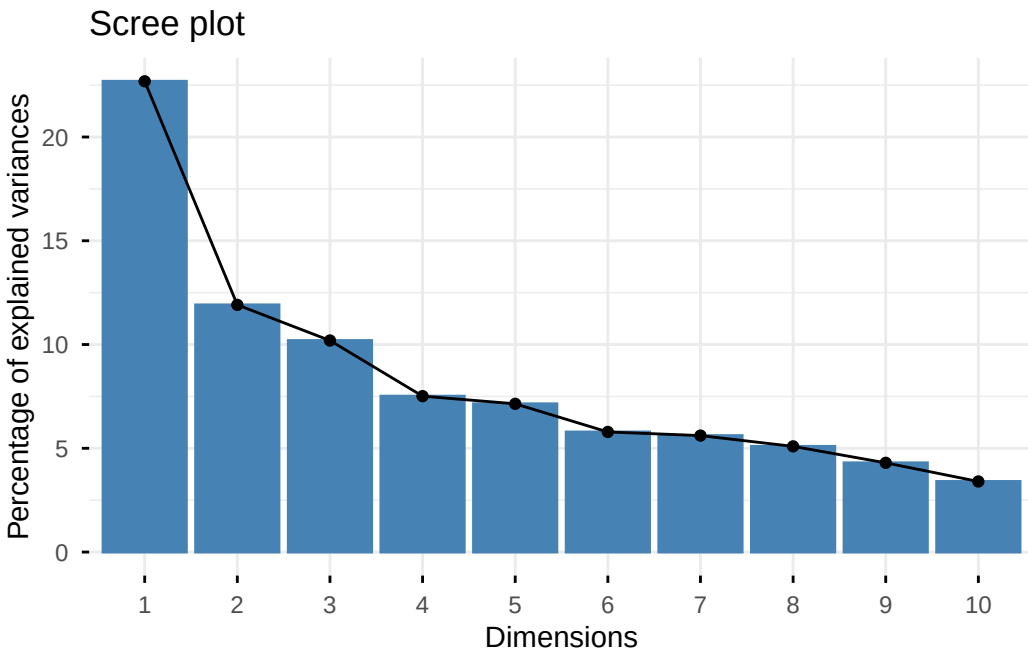
4. Principal Component Analysis and BRMS Model

Why we chose PCA

Even though we narrowed our dataset, we still had too many variables for the number of observations to produce a reliable model. To deal with this, we used Principal Component Analysis. PCA is a dimensionality reduction technique used to simplify complex datasets by transforming potentially correlated variables (e.g., kelp stipe density, holdfast cover, macroalgal abundance) into a smaller set of variables, called principal components.

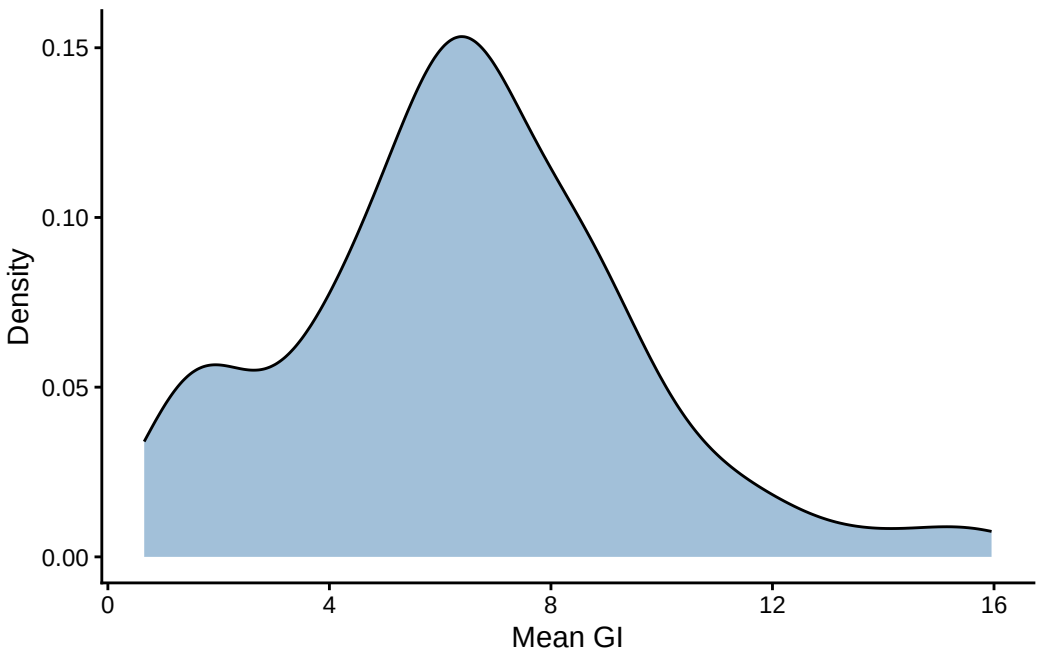
Step 1: PCA

```
#run PCA: run only on predictor variables, not the response variable (mean_gi)
#standardize variables
pca <- prcomp(predictors2 %>%
                dplyr::select(-mean_gi),
                scale. = TRUE)

#visualize
fviz_eig(pca)
```

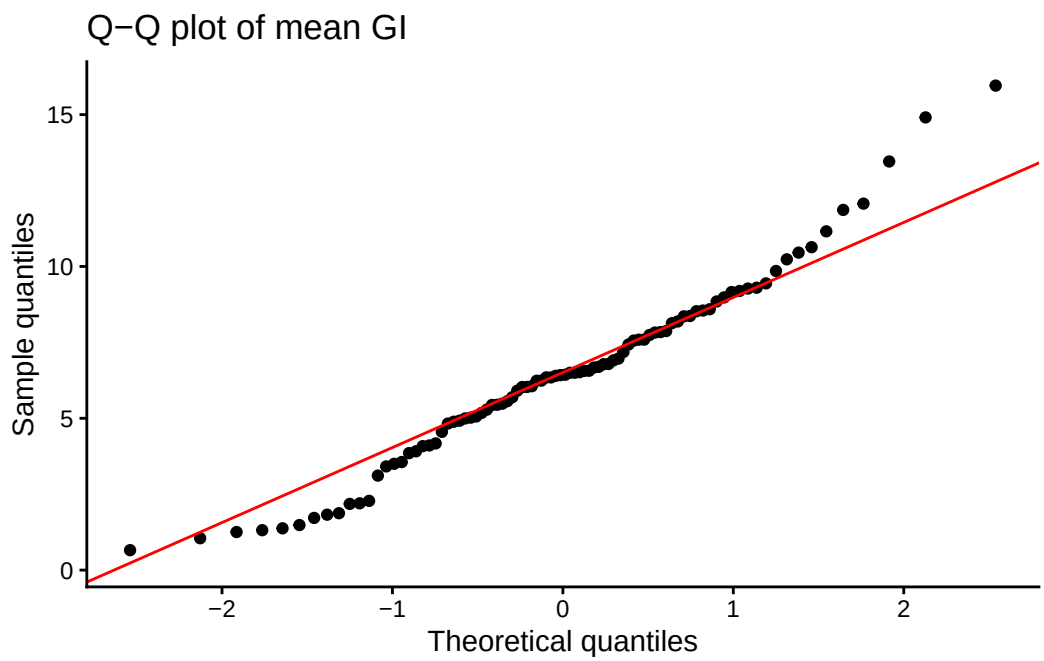


Step 2: BRM Model Setup

```
#create matrix of raw PC scores (reference scree plot)
pc_scores <- as.data.frame(pca$x[, 1:10])

#add response variable (mean_gi) into new dataframe
pc_data <- pc_scores %>%
              mutate(mean_gi = predictors2$mean_gi)

#set priors for brm using PCA
priors_pca <- c(
  prior(normal(0, 1), class = "b"),
  prior(normal(6.5, 2), class = "Intercept"), #mean mean_gi is ~6.5
  prior(student_t(3, 0, 2), class = "sigma"))

#density plot to check for normality of the data
ggplot(pc_data, aes(x = mean_gi)) +
  geom_density(fill = "steelblue", alpha = 0.5) +
  labs(x = "Mean GI", y = "Density") +
  theme_classic()
```




```
#Q-Q plot as another way to check for normality
ggplot(pc_data, aes(sample = mean_gi)) +
  stat_qq() +
  stat_qq_line(color = "red") +
  labs(x = "Theoretical quantiles",
       y = "Sample quantiles",
       title = "Q-Q plot of mean GI") +
  theme_classic()
```



Step 3: Model Selection

After checking the data's distribution and running the gaussian brm model, we wanted to see if there may be a better distribution family because the distribution is skewed. We then used the posterior predictive checks with pp_check() to visualize the model fit. Then, we re-ran the brm model with only the significant principal components (PCs) from the first gaussian model and used the same process for a skew-normal model. Comparing the model_pca_gaus with the model_pca_skew using loo_compare(), the skew-normal distribution is a better fit for predictive power.

```
#run brm model with default normal/gaussian distribution
model_pca <- brm(mean_gi ~ .,
                 data = pc_data,
                 family = gaussian(),
                 prior = priors_pca,
                 chains = 4, iter = 4000, warmup = 2000,
                 cores = 4)
```

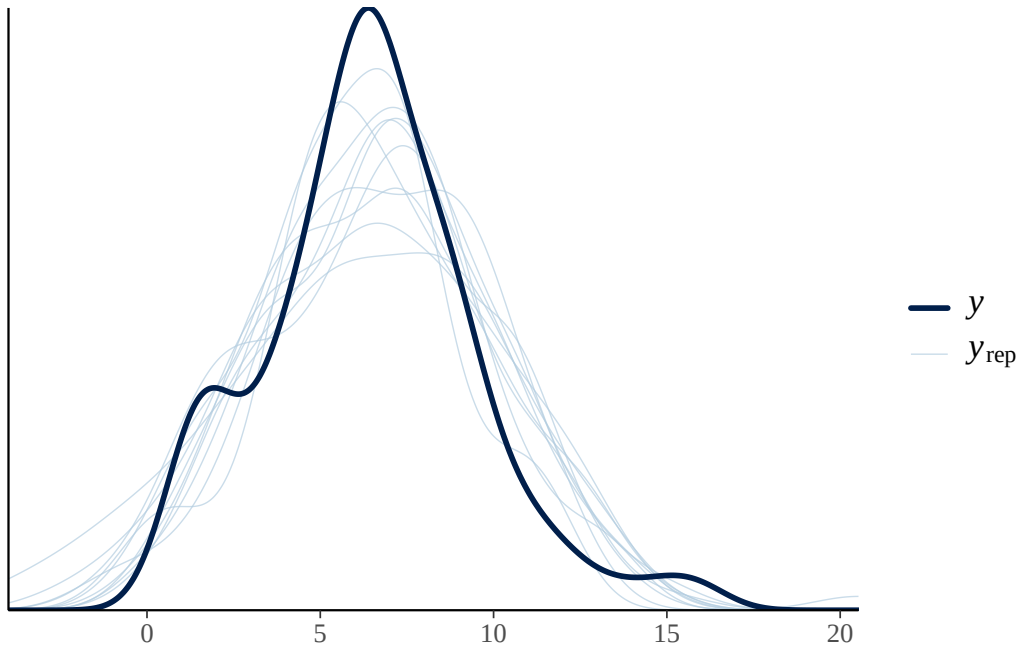
```
summary(model_pca) #PC1, PC2, and PC3 are significant!
```

```
 Family: gaussian
```

```
  Links: mu = identity
Formula: mean_gi ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6 + PC7 + PC8 + PC9 + PC10
   Data: pc_data (Number of observations: 90)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     6.44      0.26     5.94     6.93 1.00    17214     5455
PC1           0.82      0.12     0.58     1.06 1.00    16438     6044
PC2          -0.44      0.17    -0.77    -0.11 1.00    17782     6391
PC3           0.45      0.18     0.09     0.81 1.00    16475     5975
PC4           0.08      0.21    -0.33     0.50 1.00    17856     6406
PC5           0.04      0.21    -0.39     0.45 1.00    17685     5725
PC6           0.08      0.24    -0.38     0.53 1.00    15873     6064
PC7          -0.26      0.24    -0.74     0.21 1.00    18506     6372
PC8          -0.07      0.25    -0.57     0.43 1.00    15748     5572
PC9          -0.13      0.27    -0.66     0.41 1.00    18200     6275
PC10          0.17      0.30    -0.41     0.76 1.00    17296     5895

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     2.43      0.19     2.09     2.84 1.00    12372     6045

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#posterior predictive check
pp_check(model_pca) #mostly fits well, doesn't account for bump around x=2
```

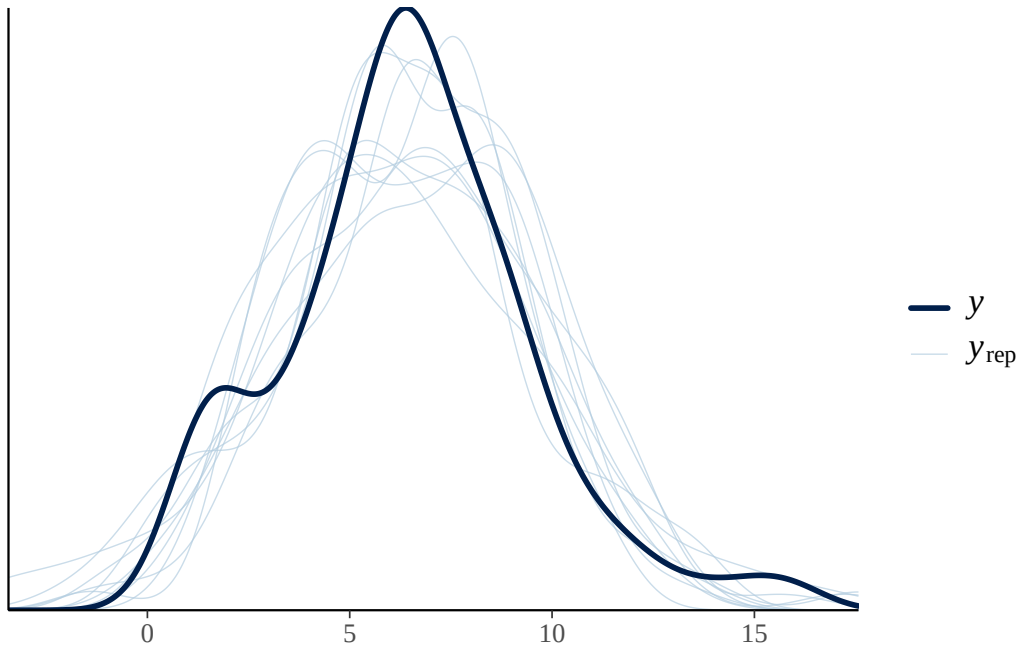
```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```




```
#re-run with only significant PCs
model_pca_gaus <- brm(mean_gi ~ PC1 + PC2 + PC3,
                                    data = pc_data,
                                    family = gaussian(),
                                    prior = priors_pca,
                                    chains = 4, iter = 4000, warmup = 2000,
                                    cores = 4)
```

```
pp_check(model_pca_gaus)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```




```r
#rerun first model with skew_normal distribution
model_pca_other <- brm(mean_gi ~ .,
                       data = pc_data,
                       family = skew_normal(), #changed to skew_normal
                       prior = priors_pca,
                       chains = 4, iter = 4000, warmup = 2000,
                       cores = 4)
```

```r
summary(model_pca_other) #again, PC1, PC2, and PC3 are significant!
```

```
 Family: skew_normal
  Links: mu = identity
Formula: mean_gi ~ PC1 + PC2 + PC3 + PC4 + PC5 + PC6 + PC7 + PC8 + PC9 + PC10
   Data: pc_data (Number of observations: 90)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     6.49      0.24     6.04     6.99 1.00     7576     5399
PC1           0.77      0.10     0.56     0.95 1.00     7483     6106
PC2          -0.57      0.13    -0.83    -0.31 1.00     7393     6154
```

```
PC3           0.37      0.14     0.11     0.67 1.00     6736     6025
PC4           0.05      0.16    -0.27     0.35 1.00     6876     5344
PC5          -0.01      0.16    -0.30     0.32 1.00     7485     5836
PC6          -0.11      0.16    -0.42     0.21 1.00     7311     5976
PC7          -0.10      0.21    -0.52     0.30 1.00     8335     6183
PC8          -0.19      0.21    -0.62     0.23 1.00     7956     6249
PC9          -0.11      0.23    -0.57     0.34 1.00     7284     6072
PC10         -0.03      0.22    -0.46     0.43 1.00     8464     6016

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     2.32      0.19     1.99     2.75 1.00     6451     4950
alpha     5.80      2.00     2.64    10.31 1.00     5829     4337

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#final model
model_pca_skew <- brm(mean_gi ~ PC1 + PC2 + PC3,  # only significant PCs
                    data = pc_data,
                    family = skew_normal(),
                    prior = priors_pca,
                    chains = 4, iter = 4000, warmup = 2000,
                    cores = 4)
```

```r
summary(model_pca_skew)
```

```
 Family: skew_normal
  Links: mu = identity
Formula: mean_gi ~ PC1 + PC2 + PC3
   Data: pc_data (Number of observations: 90)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     6.49      0.23     6.06     6.98 1.00     5372     5300
PC1           0.80      0.09     0.61     0.96 1.00     7745     5568
PC2          -0.59      0.12    -0.83    -0.34 1.00     8269     6127
PC3           0.37      0.14     0.11     0.67 1.00     7615     5660
```

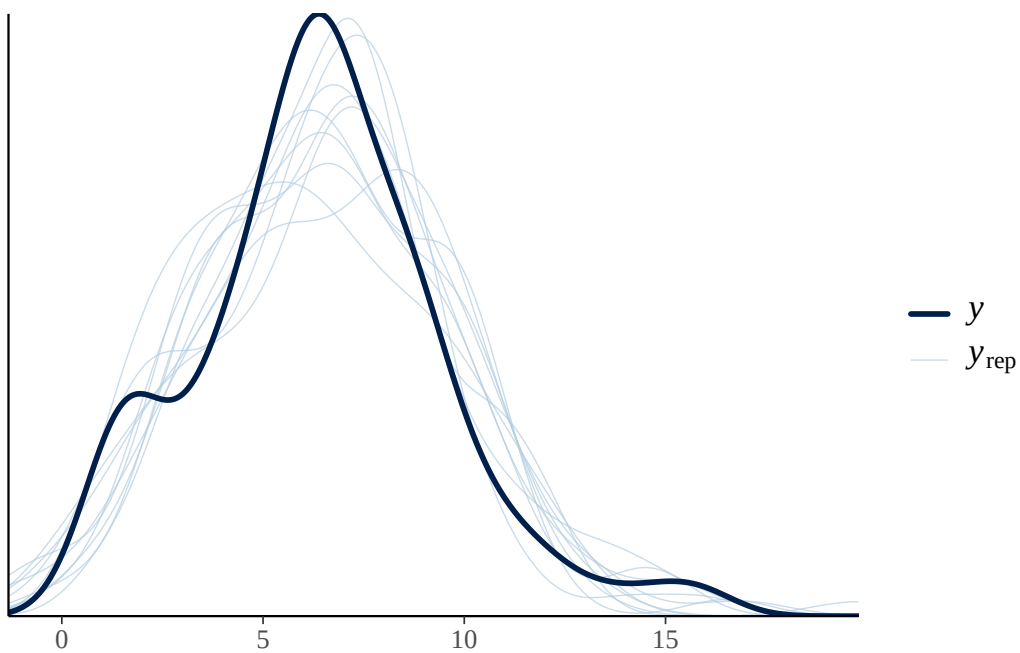
```
Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     2.27      0.18     1.95     2.67 1.00     5096     5518
alpha     5.82      1.85     2.90    10.05 1.00     6825     4957

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
bayes_R2(model_pca_skew) #check R^2
```

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.4093948 0.04211817 0.3222001 0.4890872
```

```r
#posterior predictive check
pp_check(model_pca_skew)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



```r
#compare gaussian and skew_normal models --> model_pca_skew seems to be the better model!
loo_gaussian <- loo(model_pca_gaus, moment_match = TRUE)
loo_skew <- loo(model_pca_skew, reloo = TRUE)
```

```
No problematic observations found. Returning the original 'loo' object.
```

```r
loo_compare(loo_gaussian, loo_skew)
```

```
               elpd_diff se_diff
model_pca_skew   0.0       0.0
model_pca_gaus -12.0       2.8
```

5. Interpreting and Visualizing Model

Now that we have our model, we can start to interpret and visualize it. Starting with the PCs, we created a for loop to print the variables associated with each significant PC.

PC Interpretations

PC1, PC2, and PC3 were all significant predictors of purple urchin gonad index in the Bayesian skew-normal model (all 95% CIs excluded zero, Rhats = 1.00).

PC1 was the strongest predictor of purple urchin gonad index (= 0.80, 95% CI: 0.62–0.96) and represented a gradient from kelp forest to barren-like habitat. Zones scoring high on PC1 were characterized by greater *Macrocystis* holdfast cover, stipe density, macro-algal abundance, and concealed purple urchins (meaning urchins were eating drift kelp rather than live kelp, therefore not contributing to kelp loss). In contrast, zones scoring low on PC1 were characterized by higher crustose-coralline algae cover, marine snail density, exposed red and purple urchin densities, and lower substrate relief, which can be indicators of degraded, barren-like habitat. The positive model coefficient indicates that urchins at kelp forest sites had significantly higher gonad index than those at barren-like sites, consistent with drift kelp being the primary food source supporting urchin gonad development.

PC2 had a negative effect (= -0.58, 95% CI: -0.82– -0.35), suggesting that sites with higher red urchin density, *Corynactis*, and barnacle cover were associated with lower gonad index compared to sites with articulated coralline algae and *Desmarestia*. PC3 had a moderate positive effect (= 0.37, 95% CI: 0.13–0.67), where sites with greater algal diversity were associated with higher gonad index relative to sites dominated by *Laminaria* stumps, purple urchin densities, and dead kelp holdfasts.

Together, these three habitat gradients explained 41% of variation in purple urchin gonad index (Bayesian $R^2 = 0.41$, 95% CI: 0.32–0.49).

Also, PC1, PC2, and PC3 captured 23%, 12%, and 10% of the total variation in habitat conditions (variables) respectively, together summarizing 45% of the overall habitat structure across sites.

```
#what variables are included in PC1, PC2, and PC3?
for(pc in c("PC1","PC2","PC3")) { #starts a for loop
  cat("\n---", pc, "---\n") #syntax for output: headers for each PC
  cat("TOP POSITIVE:\n") #syntax for output: label for the variables with  largest positive l
  print(head(sort(pca$rotation[,pc], decreasing = TRUE), 5)) #sorts loadings from largest to
  cat("TOP NEGATIVE:\n") #syntax for output: label for the variables with  largest negative l
  print(head(sort(pca$rotation[,pc], decreasing = FALSE), 5))} #sorts loadings from largest t
```

```
--- PC1 ---
TOP POSITIVE:
     cov_mac_holdfast_live     macro_stipe_density_m2
                 0.3680964                  0.3574166
         n_macro_plants_m2  purple_urchin_conceiledm2
                 0.3373214                  0.2761511
cov_dead_kelp_holdfast_any
                 0.2658364
TOP NEGATIVE:
 cov_crustose_coralline           marine_snails    red_urchin_densitym2
             -0.3082244              -0.2965373              -0.2233733
              relief_cm purple_urchin_densitym2
             -0.1767785              -0.1667453

--- PC2 ---
TOP POSITIVE:
cov_corynactis_californica       red_urchin_densitym2
                 0.3731614                  0.3651667
              cov_barnacle              cov_bare_sand
                 0.2918554                  0.2742160
         n_macro_plants_m2
                 0.2156306
TOP NEGATIVE:
cov_articulated_coralline       cov_desmarestia_spp purple_urchin_conceiledm2
               -0.3697764                -0.2912846                -0.2783927
            marine_snails                 relief_cm
               -0.2404735                -0.2300027
```

```
--- PC3 ---
TOP POSITIVE:
cov_articulated_coralline             cov_bare_sand         n_macro_plants_m2
                0.2013814                 0.1504267                 0.1436898
      cov_desmarestia_spp      cov_dictyoneurum_spp
                0.1365287                 0.1265713
TOP NEGATIVE:
          density_lamstump    purple_urchin_densitym2
                -0.4460719                 -0.4396316
cov_dead_kelp_holdfast_any  purple_urchin_conceiledm2
                -0.3414915                 -0.3259312
      red_urchin_densitym2
                -0.2715089
```

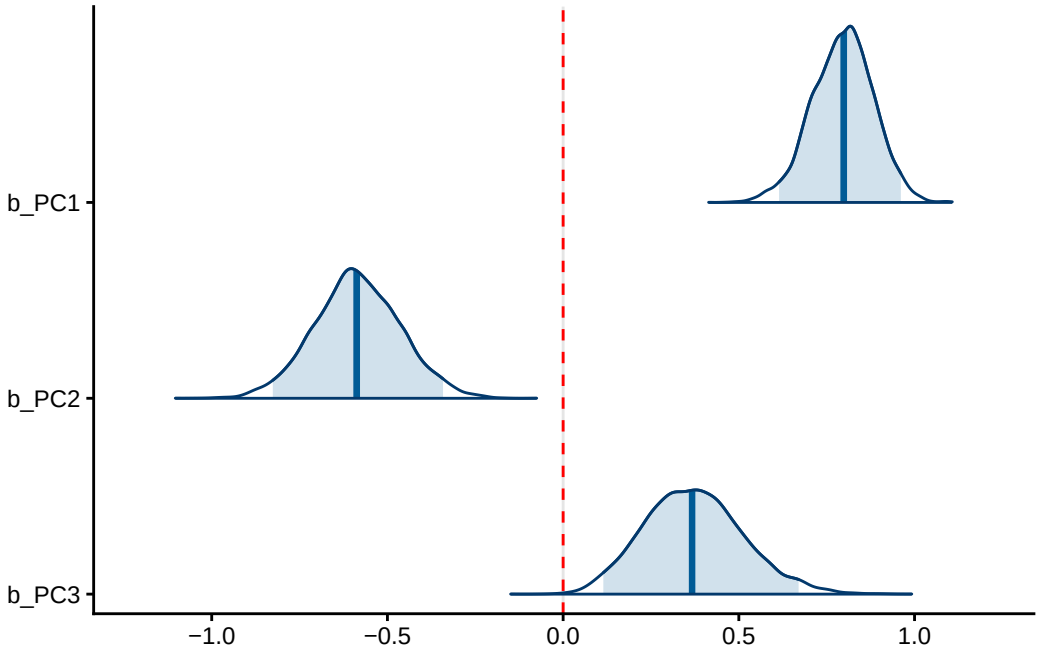
```
# % habitat structure variance for each PC
summary(pca)$importance[2, 1:3] * 100 # % variance for each PC
```

```
   PC1    PC2    PC3
22.686 11.911 10.193
```

Visualizations

The coefficient plot visually presents the three significant PCs and their relative effects on mean GI (basically visualizes the summary table). AS we can see, PC1 is furthest from x=0, meaning it has the highest effect (and positive).

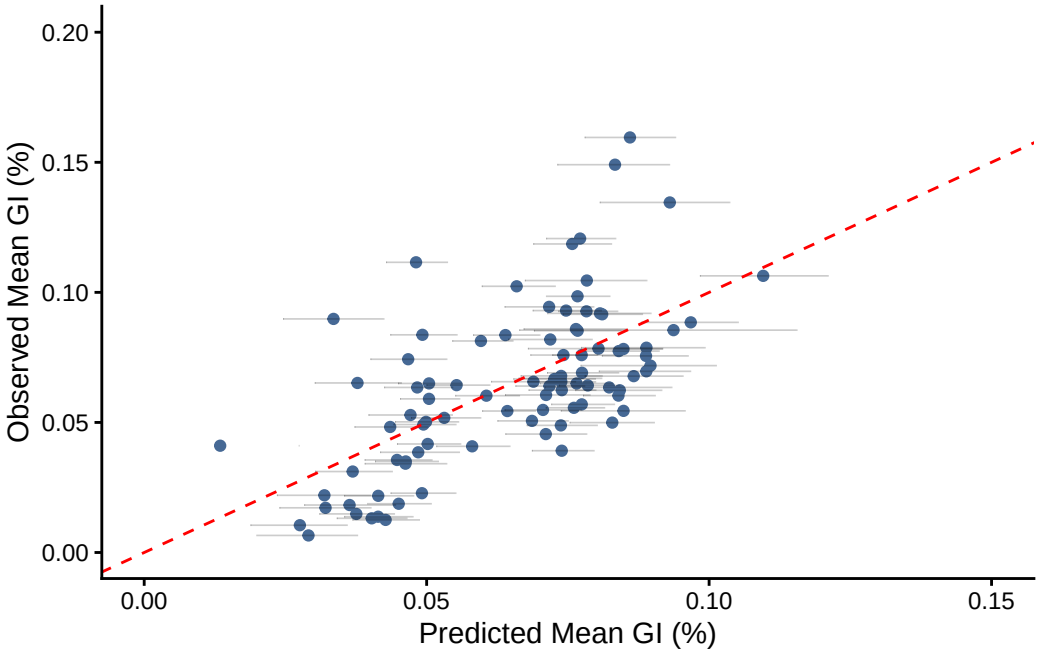
```
#density curves of each PC showing posterior distributions
mcmc_areas(model_pca_skew,
           pars = vars(starts_with("b_"), -b_Intercept), #plot regression coefficients and ex
           prob = 0.95, #shade the 95% credible interval
           prob_outer = 1.0) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "red") + #dotted line at x=0 for vi
  theme_classic()
```



To show how well the model can predict gonad index, we extracted the posterior mean predictions (fitted values) and 95% credible intervals from the model for each observation. We then plot the predictions against the observed mean gonad index values. Each point represents one survey zone within a site, with horizontal error bars showing prediction uncertainty, and the red dashed line representing the ideal predictions. The model does a fairly okay job at predicting mean GI, but because our model accounts for only about 41% of variation in mean GI, the points will not be very close to the line.

```
pc_data$fitted <- fitted(model_pca_skew)[, "Estimate"] #posterior mean prediction for each ob
pc_data$lower  <- fitted(model_pca_skew)[, "Q2.5"] #lower 95% credible interval
pc_data$upper  <- fitted(model_pca_skew)[, "Q97.5"] #upper 95% credible interval

#plot observed vs predicted to see how well the three ecologically meaningful PCs predict gon
ggplot(pc_data, aes(x = fitted/100, y = mean_gi/100)) + #turn units to %
  geom_point(color = "#466995") + #each dot is one observation
  geom_errorbarh(aes(xmin = lower/100, xmax = upper/100),
                 alpha = 0.2,
                 linewidth = 0.3) + #horizontal error bars showing prediction uncertainty
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "red") + #prediction lin
  labs(x = "Predicted Mean GI (%)", y = "Observed Mean GI (%)") +
  scale_x_continuous(limits = c(0,0.15)) +
  scale_y_continuous(limits = c(0,0.2)) +
  theme_classic()
```

```
Warning: `geom_errorbarh()` was deprecated in ggplot2 4.0.0.
i Please use the `orientation` argument of `geom_errorbar()` instead.
```



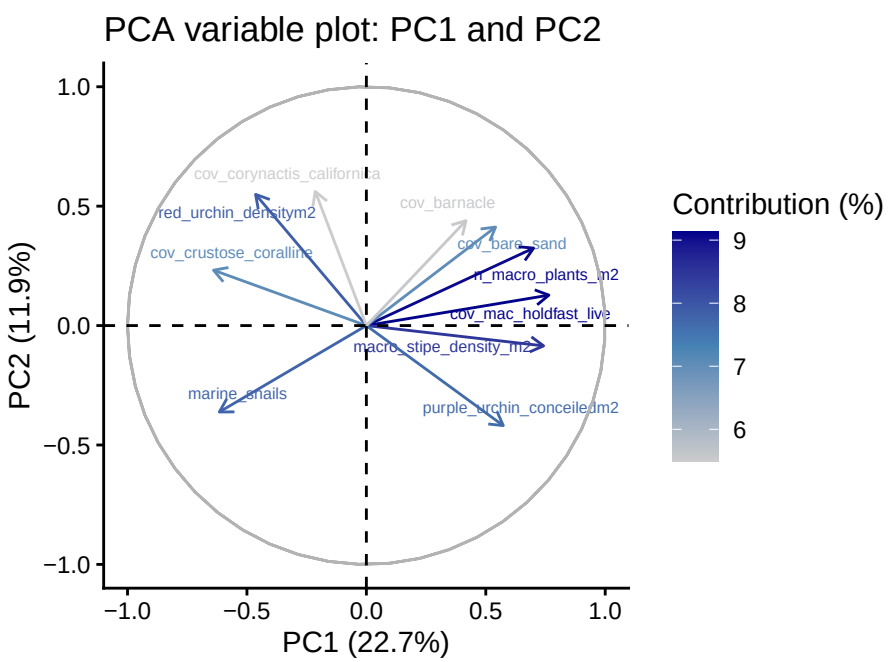
The PCA variable plot shows how the variables relate to PC1 and PC2. Arrows point to the right show varaibles correlated to each other and associated with higher mean gonad index (PC1, = 0.80), such as # of *Macrocystis* plants, cover of live *Macrocystis* holdfasts, and concealed purple urchins. Arrows to the left indicate the opposite, and are negatively correlated to GI. Arrows pointing upwards are associated with lower mean gonad index (PC2, = -0.58), such as bare sand cover, crustose-coralline algae cover, and red urchin density. The gradient of colors represents the strength of contribution, so dark blue variables/arrows are the most influential.

The three individual plots are marginal effect plots, showing the relationships between each PC and mean GI. This visualizes each PCs effect on mean GI. As we can see, the observed points are scattered quite a bit in each plot, but the points follow the line best in the PC1 plot.

```
#plotting PC1 and PC2
fviz_pca_var(pca, axes = c(1, 2), #specify which PCs to plot
             select.var = list(contrib = 10),  #only top # contributors for clear visuals
             col.var = "contrib",
             repel = TRUE,
               labelsize = 2,
```

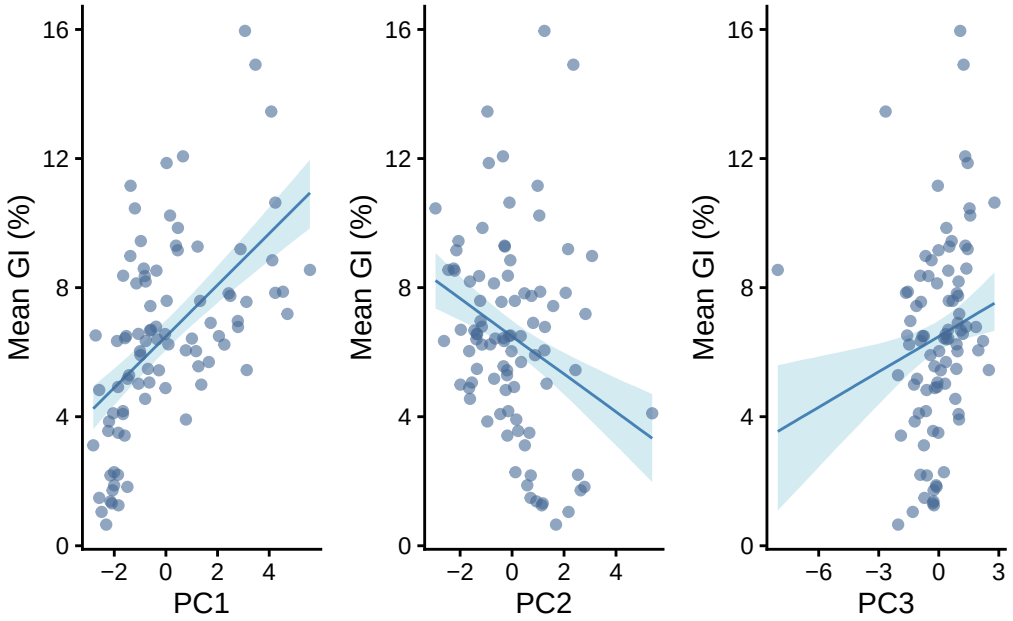
```
            gradient.cols = c("grey80", "steelblue", "darkblue")) +
  labs(title = "PCA variable plot: PC1 and PC2",
       x = "PC1 (22.7%)",
       y = "PC2 (11.9%)",
       color = "Contribution (%)")+
  theme_classic()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
i The deprecated feature was likely used in the ggpubr package.
  Please report the issue at <https://github.com/kassambara/ggpubr/issues>.

Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the factoextra package.
  Please report the issue at <https://github.com/kassambara/factoextra/issues>.
```




```
#Rework data to plot PCs separately
ce <- conditional_effects(model_pca_skew)
```

```r
# Extract the data from conditional effects
ce_pc1 <- as.data.frame(ce[["PC1"]])
ce_pc2 <- as.data.frame(ce[["PC2"]])
ce_pc3 <- as.data.frame(ce[["PC3"]])

# Build plots from scratch
p1 <- ggplot(ce_pc1, aes(x = PC1, y = estimate__)) +
  geom_ribbon(aes(ymin = lower__, ymax = upper__),
              fill = "lightblue", alpha = 0.5) +
  geom_line(color = "steelblue") +
  geom_point(data = pc_data, aes(x = PC1, y = mean_gi),
             color = "#466995", alpha = 0.6) +
  labs(x = "PC1",
       y = "Mean GI (%)") +
  theme_classic()

p2 <- ggplot(ce_pc2, aes(x = PC2, y = estimate__)) +
  geom_ribbon(aes(ymin = lower__, ymax = upper__),
              fill = "lightblue", alpha = 0.5) +
  geom_line(color = "steelblue") +
  geom_point(data = pc_data, aes(x = PC2, y = mean_gi),
             color = "#466995", alpha = 0.6) +
  labs(x = "PC2",
       y = "Mean GI (%)") +
  theme_classic()

p3 <- ggplot(ce_pc3, aes(x = PC3, y = estimate__)) +
  geom_ribbon(aes(ymin = lower__, ymax = upper__),
              fill = "lightblue", alpha = 0.5) +
  geom_line(color = "steelblue") +
  geom_point(data = pc_data, aes(x = PC3, y = mean_gi),
             color = "#466995", alpha = 0.6) +
  labs(x = "PC3",
       y = "Mean GI (%)") +
  theme_classic()

# Combine
p1 + p2 + p3
```



6. Conclusion

This project demonstrated that habitat structure is a significant predictor of purple urchin gonad index across survey sites in central California, with three principal habitat gradients collectively explaining 41% of variation in gonad condition (Bayesian $R^2 = 0.41$, 95% CI: 0.32–0.49). PC1 (representing kelp forest structure) emerged as the strongest predictor, with sites characterized by greater *Macrocystis* holdfast cover, stipe density, and macroalgal plant abundance associated with significantly higher gonad index. PC2 had a significant negative effect, indicating that exposed, barren-like sites with high red urchin density and low structural complexity were associated with poorer gonad condition. PC3 had a moderate positive effect, suggesting that sites with greater macroalgal diversity supported better urchin condition relative to sites dominated by purple urchins and dead kelp remnants. Together, these findings suggest that gonad condition may reflect the broader ecological state of a site, and that habitat-based variables may serve as useful indicators for predicting urchin condition. For restoration practitioners, these results highlight the importance of restoring not just urchin density but the broader habitat conditions that support urchin condition, and suggest that sites with intact kelp structure and high algal diversity are likely to support urchins with greater reproductive potential and higher gonad quality.